THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Light travels in 1

(a) zigzag paths **(b)** straight lines **(c)** spirals **(d)** circles

2. The bending of light when it passes from one medium to another is 1

(a) dispersion **(b)** diffraction **(c)** refraction **(d)** reflection

3. A lens is a transparent material bounded by 1

(a) four surfaces **(b)** two spherical surfaces

(c) one plane surface only **(d)** no surface

4. The bouncing back of light from a surface is called 1

(a) refraction **(b)** reflection **(c)** scattering **(d)** dispersion

5. A ray of light going from air to water bends 1

(a) randomly **(b)** towards the normal

(c) away from the normal **(d)** not at all

6. A convex lens is thicker at the 1

(a) only at one side **(b)** centre **(c)** both equally **(d)** edges

7. In reflection, the angle of incidence is equal to 1

(a) zero **(b)** the angle of reflection

(c) 90 degrees **(d)** the angle of refraction

8. The speed of light is maximum in 1

(a) diamond **(b)** vacuum **(c)** water **(d)** glass

9. A concave lens is thicker at the 1

(a) centre **(b)** edges **(c)** both equally **(d)** only at one side

10. The perpendicular drawn at the point of incidence is the 1

(a) axis **(b)** normal **(c)** pole **(d)** tangent

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is reflection of light? 2

12. Which mirror gives a diminished image always? 2

13. Which has more refractive index, glass or water? 2

14. State the nature of the image formed by a plane mirror. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Name the type of mirror used for 3
(a) shaving
(b) torch
(c) rear-view of vehicle.

16. Find the refractive index of diamond if light travels in it at 1.25×10^8 m/s. 3

17. State the lens formula and the formula of magnification. 3

OR

Write the laws of reflection of light. 3

18. Why does a ray of light bend when it enters water? 3

OR

What is meant by the term 'principal axis'? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the images formed by a concave mirror when the object is 5
(a) at infinity
(b) at C
(c) between P and F.

OR

An object is placed 40 cm from a concave mirror of focal length 20 cm. Find the image position and nature. 5

20. Explain why convex mirrors are used as rear-view mirrors. 5

OR

Find the image distance and magnification when an object is placed 20 cm from a convex lens of focal length 10 cm. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND